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# Chlorophyll Determination

## Determination of Chlorophyll Concentration in Biomatter using Spectrophotometry

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# 1 Background

## 1.1 Chlorophyll

Chlorophyll is a green pigment found in plants and some bacteria. It is responsible for the green color of plants and is essential for photosynthesis. It consists of two main types: Chlorophyll a ( $\text{Chl}_a$ ) and Chlorophyll b ( $\text{Chl}_b$ ).

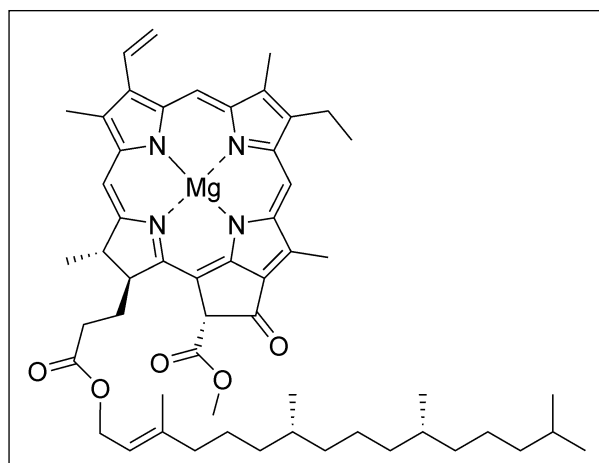


Figure 1: Chlorophyll A structure <sup>[1]</sup>

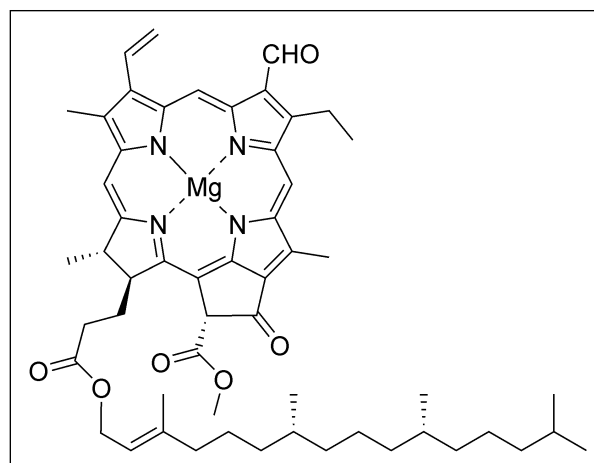


Figure 2: Chlorophyll B structure <sup>[2]</sup>

### 1.1.1 Solubility

Chlorophyll has a long fatty chain called phytol ( $\text{C}_{20}\text{H}_{39}$ ) attached to it. This chain makes chlorophyll dissolve well in oils and organic solvents, but not in water. The phytol chain acts like an anchor that holds the chlorophyll in place within the plant's membrane structures.

Because of this fatty chain, organic solvents like acetone, methanol, or ethanol are needed to extract chlorophyll from plant tissues. 80% acetone (mixed with 20% water) is commonly used because the small amount of water helps break apart the protein structures that hold the chlorophyll. <sup>[3], [4]</sup> 100% acetone is very suitable for leave extracts. <sup>[3]</sup>

### 1.1.2 Solvatochromism

The solvent effect, known as "Solvatochromism", is a phenomenon that occurs when the absorption spectrum of a compound changes between different solvents used. This solvent-dependent effect applies to chlorophyll and arises from the interaction between the solvent molecules and the chlorophyll's  $\pi$ -electron cloud <sup>[5], [6]</sup>.

Consequently, absorption coefficients derived for methanol cannot be used for acetone extracts, and coefficients for 100% acetone cannot be used for 80% acetone. The specific equations discussed in Section 5.2 are strictly valid only for 100% Acetone extracts. <sup>[3], [5]</sup>

### 1.1.3 Photophysics and structural distinctions

Chlorophylls are Mg-coordinated chlorins (porphyrin derivatives) with an extended conjugated  $\pi$ -system that gives rise to two dominant absorption regions: an intense Soret band in the blue (430–460 nm) and a weaker Q band in the red (640–670 nm). The weak absorption in the green explains the perceived color of foliage <sup>[5]</sup>.

The only structural difference between  $\text{Chl}_a$  and  $\text{Chl}_b$  at C-7 (methyl vs. formyl) shifts the spectra: in 80% acetone,  $\text{Chl}_a$  peaks near 664nm and  $\text{Chl}_b$  near 647nm, enabling simultaneous

quantification in mixtures<sup>[1], [2]</sup>. The hydrophobic phytol tail anchors chlorophylls in thylakoid membranes and dictates solubility in organic solvents<sup>[4]</sup>.

## 1.2 Spectrophotometry

### 1.2.1 Historical evolution of quantification

From the first isolations and naming of the “green matter” (Pelletier & Caventou) and early structural elucidation (Willstätter & Stoll) to modern spectrophotometry, the methodology evolved substantially<sup>[7], [8]</sup>. Early quantitative work (e.g. Arnon’s 80% acetone method at 663/645 nm) established practical lab protocols but used coefficients limited by instrument resolution<sup>[9]</sup>. Later high-resolution work by Lichtenthaler, Wellburn and Porra refined solvent-specific extinction coefficients and simultaneous equations for 80% acetone, methanol, and other solvents, which underpin the factors we use in this report<sup>[3], [10], [11]</sup>.

### 1.2.2 Beer-Lambert law

The Beer-Lambert law, shown in Equation 1, describes the correlation between the dimensionless total absorbance  $E_\lambda$ , the molar extinction coefficient  $\epsilon_\lambda$  ( $\frac{\text{l}}{\text{mol} \cdot \text{cm}}$ <sup>[12]</sup>) and concentration  $c$  ( $\frac{\text{mol}}{\text{l}}$ ) of the substance of interest in the sample, and the light path length  $d$  (cm) through the sample, where  $\lambda$  is an arbitrary wavelength of light.

$$(1) \quad E_\lambda = \epsilon_\lambda \cdot c \cdot d$$

For example  $\text{Chl}_a$  has  $\epsilon_{420} = \frac{85\,000}{\text{mol} \cdot \text{l}}$ . This however depends on a multitude of factors, such as pH, temperature or solvent, and is therefore not a constant<sup>[12]</sup>.

### 1.2.3 Mixtures and additivity

In an extract, total absorbance at a given wavelength is the sum of each pigment’s absorbance:

$$(2) \quad E_\lambda = \sum_i \epsilon_{i,\lambda} \cdot c_i \cdot d$$

### 1.2.4 Calculation of Chlorophyll Concentration

By rearranging Equation 1 we get:

$$(3) \quad c \left[ \frac{\text{mol}}{\text{l}} \right] = \frac{E_\lambda}{\epsilon_\lambda} \cdot \frac{1}{d}$$

$d$  is the path length through the sample or in other words the thickness of the cuvette. The global standard for this is  $d = 1 \text{ cm}$ <sup>[13]</sup>. Therefore we can effectively cancel out  $d$  in the equation.

While Equation 3 is a general equation yielding  $\frac{\text{mol}}{\text{l}}$ , we can convert to mass concentration  $\frac{\text{g}}{\text{l}}$  by multiplying with the molar mass  $\text{MG}_S$  of our substance  $S$  and factoring in the dilution factor  $\frac{V_E}{V_P}$ , where  $V_E$  is the volume of the extraction solvent and  $V_P$  is the volume of the biological source material.

$$(4) \quad c \left[ \frac{g}{l_E} \right] = \frac{E_\lambda}{\varepsilon_\lambda} \cdot MG_S \cdot \frac{1}{d} \Rightarrow$$

$$c \left[ \frac{g}{l_P} \right] = c \left[ \frac{g}{l_E} \right] \cdot \frac{V_E}{V_P}$$

Here  $c \left[ \frac{g}{l_E} \right]$  is the mass concentration of the substance S in the extraction solvent and  $c \left[ \frac{g}{l_P} \right]$  is the mass concentration of the substance S per volume of the biological source material.

**Note:** We can use the same approach as for removing the dilution factor  $\frac{V_E}{V_P}$  to convert from  $\frac{g}{l_E}$  to  $\frac{mg}{g_P}$  by multiplying with the dilution factor  $\frac{V_E}{M_P}$ , as shown in Equation 5.

$$(5) \quad c \left[ \frac{g}{g_P} \right] = c \left[ \frac{g}{l_E} \right] \cdot \frac{V_E}{M_P} \cdot 1000$$

Where  $M_P$  is the mass of the biological source material in g, and  $c \left[ \frac{g}{g_P} \right]$  is the mass of the substance S per weight of biological source material. This is often used to determine the concentration in fresh weight.

## 2 Execution

## 3 Methodological Notes

- Grinding with quartz/sea sand improves mechanical rupture of cell walls and thylakoids, increasing extraction efficiency <sup>[4]</sup>.
- Add a small amount of  $\text{CaCO}_3$  during grinding to neutralize organic acids and prevent pheophytinization (loss of  $\text{Mg}^{2+}$  and spectral shifts) <sup>[4]</sup>.
- Work in dim light and keep samples cool to limit photo-oxidation and pigment degradation <sup>[4]</sup>.
- 80% acetone balances membrane disruption with protein hydration and has well-defined coefficients for simultaneous equations; use solvent-specific equations only for their matching solvent <sup>[3]</sup>.
- Clarify extracts (preferably by centrifugation); if available, correct for scatter with  $E_{750}$  subtraction before pigment calculations <sup>[3]</sup>.

## 4 Instrument limits and alternative solvents

- Spectral bandwidth and stray light can flatten sharp peaks and bias  $E_\lambda$ ; low-resolution colorimeters may underestimate chlorophyll if high-resolution coefficients are used <sup>[10]</sup>.
- Alternative solvents exist: DMF (high efficiency, but hazardous) and methanol (efficient yet can promote allomerization). Use solvent-specific coefficients/equations accordingly <sup>[10], [11]</sup>.

## 5 Results

We performed various analysis tasks in order to gain a better understanding of the data and the results. These were done in isolation to avoid any confounding factors.

We decided to take a detailed full spectrum sample using a mixture of brussels sprouts and maggi herbs <sup>1</sup>. We will use this sample for all single sample analysis, especially in Section 5.2 and Section 5.3. We refer to this sample as “our’s” going forward.

### 5.1 Spectral Analysis

The full spectrum data was derived by sampling absorption rates at 10nm intervals from the spectrophotometer’s automatic full spectrum curve. Results are shown in Figure 4.

#### 5.1.1 Reference spectrum

We compare our full spectrum data with the reference spectra of chlorophyll a and chlorophyll b in diethyl ether. This is used for reference even though we used acetone in our experiment, because we failed to find other publically available full spectrum reference data.

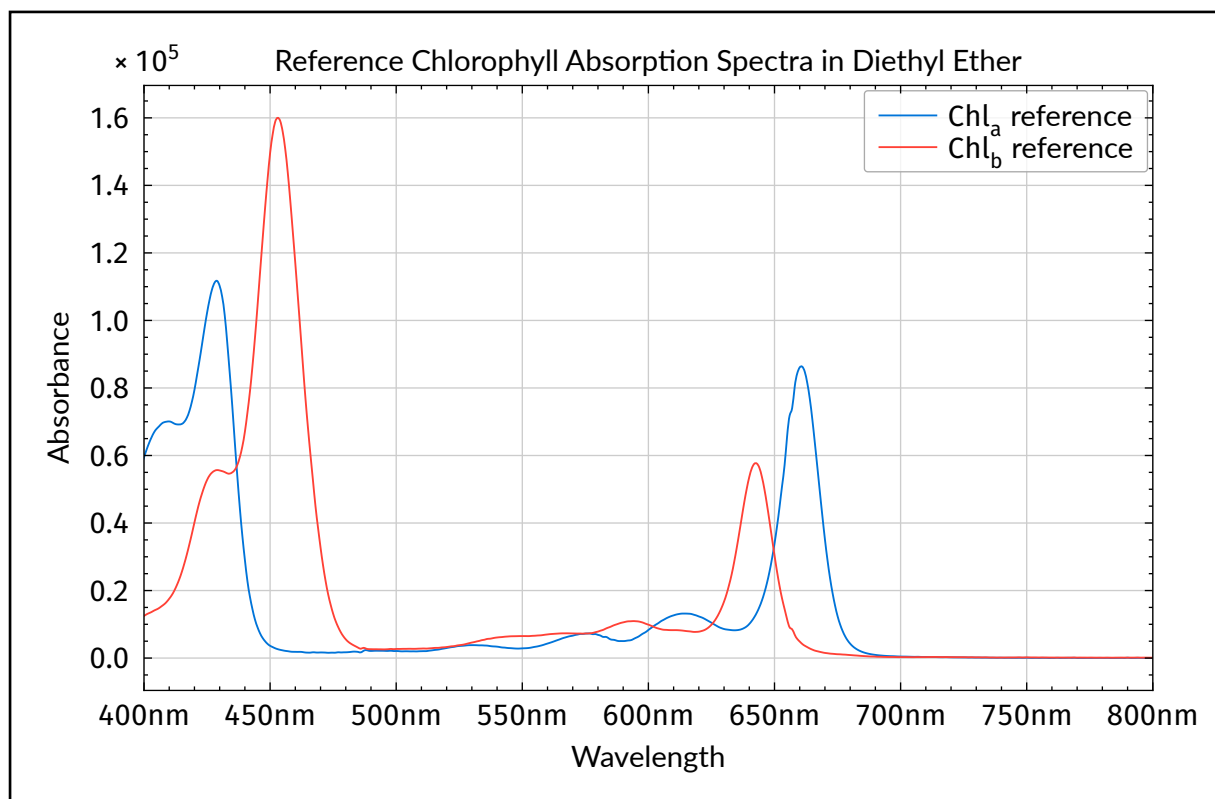


Figure 3: Reference Chlorophyll Absorption Spectra in Diethyl Ether <sup>[1], [2]</sup>. Blue shows Chl<sub>a</sub> and red shows Chl<sub>b</sub>.

In Figure 3 we can observe the blue spectrum peaks at ~ 430nm for Chl<sub>a</sub> and ~ 452nm for Chl<sub>b</sub>. The red spectrum peaks at ~ 660nm for Chl<sub>a</sub> and ~ 642nm for Chl<sub>b</sub>.

These match the characteristics of the chlorophylls absorption spectrum in diethyl ether. We can observe the spectrum line of Chl<sub>a</sub> buckles when crossing the line of Chl<sub>b</sub> in the blue region, which is also characteristic for both chlorophyll variants. <sup>[5]</sup>

<sup>1</sup>File group 1, subgroup 6 from the results sheet

Therefore, we conclude the found dataset to be a viable and accurate source of reference.

### 5.1.2 Sample spectrum

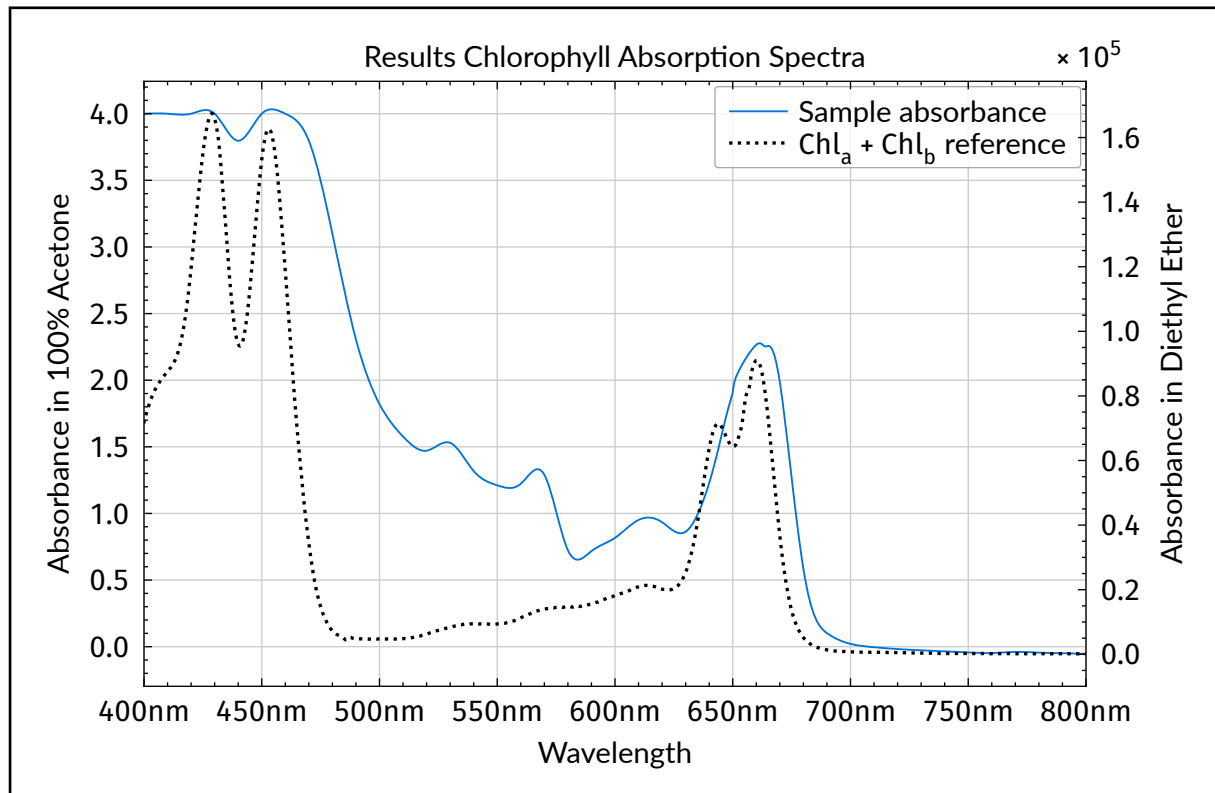


Figure 4: Absorption Spectrum of the Sample (shown in blue) and combined  $\text{Chl}_a + \text{Chl}_b$  Reference from Figure 3 (shown in dotted black)

The sample spectrum shows the absorption of the total, non-isolated chlorophylls ( $\text{Chl}_a + \text{Chl}_b$ ) in the sample. We can observe three peaks: two in the blue region at  $\sim 430\text{nm}$  and  $\sim 445\text{nm}$  with one in the red region at  $\sim 660\text{nm}$ .

We can observe similar peaks as in the reference spectrum, but with slight distortions and more or less pronounced features. This is because the sample consists of total, non-isolated chlorophylls ( $\text{Chl}_a + \text{Chl}_b$ ) and possibly other pigments dissolved in acetone. Furthermore, as mentioned in Section 1.1.2, shifts are to be expected when comparing our sample with diethyl ether dissolved reference<sup>[5]</sup>. Despite this, one can even spot fine features in our spectrum. For example the valley at the  $\sim 650\text{nm}$  wavelength that is clearly visible in the reference spectrum is present in our data in form of a small bump.

The most claring difference compared to reference is the samples gradually increasing absorption in the green spectrum (read from right/red to left/blue), whereas according to the reference the absorption should slightly decrease. One possible explanation for this is the presence of other pigments in the sample that absorb in the green spectrum, such as carotenoids<sup>[4], [5]</sup>. Another plausible explanation is incomplete calibration: Calibration did not happen across the entire spectrum but only once on the higher end of the wavelength. The level of unfamiliarity with the spectrophotometer and the spectrometer itself might have contributed to this.

## 5.2 Chlorophyll Concentration with Instructor's Formula

The instructor provided us with a formula to calculate the chlorophyll concentration:

$$\begin{aligned}
 (6) \quad c_a \left[ \frac{\text{mg}}{\text{l}} \right] &= 11.78 E_{664} - 2.29 E_{647} \\
 c_b \left[ \frac{\text{mg}}{\text{l}} \right] &= 20.05 E_{647} - 4.77 E_{664} \\
 c_{\text{total}} &= c_a + c_b = 27.8 E_{652}
 \end{aligned}$$

### 5.2.0.1 Origin of the coefficients (80% acetone)

Using solvent-specific specific absorption coefficients (80% acetone) for  $\text{Chl}_a$  and  $\text{Chl}_b$  at 664nm and 647nm (e.g.  $\alpha_{a,664} \approx 89.0$ ,  $\alpha_{b,664} \approx 10.2$ ,  $\alpha_{a,647} \approx 21.2$ ,  $\alpha_{b,647} \approx 52.3$  in  $\text{L} \cdot \text{g}^{-1} \cdot \text{cm}^{-1}$ ) and solving the two-equation system yields:

$\text{Chl}_a \left[ \frac{\text{mg}}{\text{l}} \right] = 11.78 E_{664} - 2.29 E_{647}$ ,  $\text{Chl}_b \left[ \frac{\text{mg}}{\text{l}} \right] = 20.05 E_{647} - 4.77 E_{664}$ . These correspond to the factors we use in this report and are consistent with refined equations reported in the literature for 80% acetone <sup>[3], [10], [11]</sup>.

Plugging in the concrete values, shown in Table 1, for our sample we get:

$$\begin{aligned}
 c_a \left[ \frac{\text{mg}}{\text{l}} \right] &= 11.78 \cdot 2.253 - 2.29 \cdot 1.715 = \mathbf{22.61299} \\
 c_b \left[ \frac{\text{mg}}{\text{l}} \right] &= 20.05 \cdot 1.715 - 4.77 \cdot 2.253 = \mathbf{23.63894} \\
 c_{\text{total}} \left[ \frac{\text{mg}}{\text{l}} \right] &= c_a + c_b = 22.61299 + 23.63894 = \mathbf{46.25193} \\
 &= 27.8 E_{652} = 27.8 \cdot 2.049 = \mathbf{56.9622}
 \end{aligned}$$

| Wavelength | Absorbance |
|------------|------------|
| 664        | 2.253      |
| 647        | 1.715      |
| 652        | 2.049      |

Table 1: Wavelengths and absorbance of the results from our sample needed for calculating the chlorophyll concentration using the given formulas

A quick sanity check with the Gemini LLM for the plausability of these values was positive. Lab extracts have a common range of  $10 - 30 \frac{\text{mg}}{\text{l}}$ . This aligns with our values.

We can see a significant difference between the total result using the standalone absorbance factor and the addition of  $\text{Chl}_a$  and  $\text{Chl}_b$ , with an absolute difference of  $10.7103 \frac{\text{mg}}{\text{l}}$  and a relative difference of 18.8 %.

### 5.2.1 Interpretation of the difference

It proves difficult to reason for a possible explanation for this discrepancy without knowing where the given formulas originate from and how they were derived. Measurement or execution errors are a possibility, even though seeming unlikely due to the high difference.

## 5.3 Chlorophyll Concentration using Beer-Lambert Law

In order to find both the  $\text{Chl}_a$  and  $\text{Chl}_b$  concentration, we need to use the Beer-Lambert law. We arrange the Equation 1 into a linear equation system to solve for both concentrations:

$$(7) \quad \begin{aligned} E_{664} &= \varepsilon_{a,664} \cdot c_a + \varepsilon_{b,664} \cdot c_b \\ E_{647} &= \varepsilon_{a,647} \cdot c_a + \varepsilon_{b,647} \cdot c_b \end{aligned}$$

Because we use the standard path length of  $d = 1\text{cm}$  we can ignore  $\frac{1}{d}$  in our calculations. Using substitution we can solve for  $c_a$  and  $c_b$ :

$$(8) \quad \begin{aligned} c_a \left[ \frac{\text{mol}}{\text{l}} \right] &= \frac{E_{664} \varepsilon_{b,647} - E_{647} \varepsilon_{a,664}}{\varepsilon_{a,664} \varepsilon_{b,647} - \varepsilon_{a,647} \varepsilon_{b,664}} \\ c_b \left[ \frac{\text{mol}}{\text{l}} \right] &= \frac{E_{647} \varepsilon_{a,664} - E_{664} \varepsilon_{b,647}}{\varepsilon_{a,664} \varepsilon_{b,647} - \varepsilon_{b,664} \varepsilon_{a,647}} \end{aligned}$$

We arrive at final formula for calculating the total chlorophyll concentration, accounting for the mass concentration and dilution factor:

$$(9) \quad \begin{aligned} c_a \left[ \frac{\text{g}}{\text{l}_E} \right] &= c_a \left[ \frac{\text{mol}}{\text{l}} \right] \cdot \text{MG}_{\text{Chl}_a} \\ c_b \left[ \frac{\text{g}}{\text{l}_E} \right] &= c_b \left[ \frac{\text{mol}}{\text{l}} \right] \cdot \text{MG}_{\text{Chl}_b} \\ c_{\text{total}} \left[ \frac{\text{g}}{\text{l}_E} \right] &= c_a \left[ \frac{\text{g}}{\text{l}_E} \right] + c_b \left[ \frac{\text{g}}{\text{l}_E} \right] \\ c_{\text{total}} \left[ \frac{\text{g}}{\text{g}_P} \right] &= c_{\text{total}} \left[ \frac{\text{g}}{\text{l}_E} \right] \cdot \frac{V_E}{M_P} \end{aligned}$$

By plugging in the values from Table 2 for this equation we arrive at the final values for the mass concentration in the extraction solvent:

$$c_a \left[ \frac{\text{mg}}{\text{l}_E} \right] = \mathbf{23.38}$$

$$c_b \left[ \frac{\text{mg}}{\text{l}_E} \right] = \mathbf{24.27}$$

$$c_{\text{total}} \left[ \frac{\text{mg}}{\text{l}_E} \right] = \mathbf{47.65}$$

Accounted for the dilution factor, the final determined mass concentration in the biological source material (fresh weight) is:

$$c_{\text{total}} \left[ \frac{\text{mg}}{\text{g}_P} \right] = c_{\text{total}} \left[ \frac{\text{mg}}{\text{l}_E} \right] \cdot \frac{V_E}{M_P} = \mathbf{0.4984}$$

|                | $\text{MG}_S$             | $\varepsilon_{647}$                         | $\varepsilon_{664}$                         | $E_{647}$ | $E_{664}$ | $V_E \div M_P$                                   |
|----------------|---------------------------|---------------------------------------------|---------------------------------------------|-----------|-----------|--------------------------------------------------|
| $\text{Chl}_a$ | 893.5 g mol <sup>-1</sup> | 17 464 l mol <sup>-1</sup> cm <sup>-1</sup> | 76 790 l mol <sup>-1</sup> cm <sup>-1</sup> | 1.715     | 2.253     | $\frac{19.98\text{ml}}{1910\text{mg}} = 0.01046$ |
| $\text{Chl}_b$ | 907.5 g mol <sup>-1</sup> | 47 040 l mol <sup>-1</sup> cm <sup>-1</sup> | 9 121 l mol <sup>-1</sup> cm <sup>-1</sup>  |           |           |                                                  |

Table 2: Extinction coefficients <sup>[14]</sup>, molecular weights, absorbance values and dilution factor needed for calculating the chlorophyll concentration using the Beer-Lambert law.

### 5.3.1 Carotenoids (optional, single extract)

When  $E_{470}$  is available (same 80% acetone extract), a widely used estimator for total carotenoids  $C_{x+c}$  corrects for chlorophyll overlap in the blue:

$$(10) \quad C_{x+c} = \frac{1000 E_{470}^* - 1.90 \text{ Chl}_a - 63.14 \text{ Chl}_b}{214}$$

with baseline-corrected  $E_{470}^* = E_{470} - E_{750}$  and  $\frac{\text{mg}}{\text{l}}$  units for  $\text{Chl}_a$  and  $\text{Chl}_b$  [3].

### 5.3.2 Comparison to instructor's formula

The values calculated using the Beer-Lambert law are in line with the values calculated using the instructor's formulas in Section 5.2. The difference is small (2.93 %) and indicates that the Beer-Lambert law and our calculations are a valid method for determining the chlorophyll concentration. Small differences can be attributed to different extinction coefficients of the chlorophylls used when precalculating the given formulas' factors.

Section 5.2.1 describes the unknown origin of the instructor-provided formulas. After performing the calculations ourselves, we can observe that once the extinction coefficients  $\epsilon$  and molecular weights MG are known for both chlorophyll types, the concentration equation reduces to simple multilinear functions with constant factors applied to the remaining variables  $E_\lambda$ . This suggests that the given formulas are simplified versions of the complete calculation chain we performed based on the Beer-Lambert law.

Interestingly enough, even though the extinction coefficients  $\epsilon$  used in our manual calculations are for 80% acetone solutions [14], the Beer-Lambert law calculations still yield results in line with the instructors formulas, despite the solvatochromism effect, stated in Section 1.1.2. The small difference between the results of the instructor's formula and the Beer-Lambert law calculations could then be explained by this and the hypothesis, that the absorption difference between 80% acetone and 100% acetone is not significant enough to cause a noticeable difference in the calculated concentration.

### 5.3.3 Checking for alignment with known concentrations

With the calculated concentration value in fresh weight for our sample, we want to check for plausability by comparing to other well known concentrations of common leafy vegetables in fresh weight.

Chlorophyll content varies widely by species, leaf age, and sunlight exposure.

- Total Chlorophyll (a+b): Generally ranges from 0.08 to 19.2 mg/g fresh weight.
- Common Leafy Vegetables:
  - Spinach: 120-150 mg/100g ( 1.2-1.5 mg/g).
  - Chard: 121 mg/100g.
  - Lettuce: 19-58 mg/100g.
  - Parsley: 5-20 µg/g.

— Gemini LLM 2025

Chlorophyll Content Comparison:

Lovage generally has higher chlorophyll concentrations than most lettuce varieties but slightly lower levels than the most dense leafy herbs like parsley.

- **Lovage (Maggi Herb):** 0.68 mg/g (68.5 mg/100g)
- **Brussels Sprouts:** 2.41 mg/g
- **Lettuce:** 0.12 - 1.0 mg/g
- **Parsley:** 2.18 - 2.23 mg/g

— Gemini LLM 2025

Using this information and normalizing to mg/g of fresh weight, the results of which are shown in Table 3, we can compare our results in fresh weight to the known concentrations of common leafy vegetables.

**Note:** Even though Gemini is no scientific source, we only use it for a basic plausability check of our own results. It can therefore be deemed a reliable enough source for this specific usecase.

The total chlorophyll concentration of our sample is 0.4984 mg/g of fresh weight. This is less than known concentrations of both brussel sprouts and maggi herbs, which are the only components of our analysed sample. This is unexpected and suggests multiple potential error sources. These could include inadequate processing/grinding with quartz sand, the fact that the material was cooled and partially frozen during preparation, incomplete extraction due to insufficient contact time with the solvent, or degradation of chlorophyll during sample handling.

Nevertheless, the results are within general ranges of our expected values, and are therefore deemed plausible. We do not suspect a great systematic error in our measurements, but rather a combination of multiple smaller errors.

## 5.4 Physiological interpretation

- Chl a/b ratio reflects light acclimation: higher in sun-exposed leaves (smaller antennae), lower in shade (larger LHCs). Typical ranges 3.0–4.0 (sun) vs. 2.0–2.5 (shade) <sup>[10]</sup>.
- Total chlorophyll ( $c_{a+b}$ ) declines under stress (chlorosis), including N or Mg limitation (biosynthetic constraints) and during senescence (nutrient remobilization) <sup>[5]</sup>.
- The “green gap”: although solutions absorb little at 550 nm, green light penetrates deeper in intact leaves and is utilized by lower chloroplast layers <sup>[5]</sup>.
- Olive-brown color and anomalously low  $Chl_a$  can indicate pheophytinization (acid-induced  $Mg^{2+}$  loss); timely processing, cooling, dim light and  $CaCO_3$  help prevent this artifact <sup>[4]</sup>.

## 5.5 Descriptive Statistics

We analyze the results of all groups in our class and compare them to our own results. We also compare different source materials with each other. Except for explicitly stated otherwise, the concentrations in fresh weight are computed using the instructor’s formula for individual chlorophyll types and adding them together, after which dilution compensation is performed. We do not use the pre-calculated concentrations from the results sheet.

| Spinach   | Chard | Lovage (Maggi Herb) | Brussels Sprouts | Lettuce  | Parsley     |
|-----------|-------|---------------------|------------------|----------|-------------|
| 1.2 - 1.5 | 1.21  | 0.68                | 2.41             | 0.12 - 1 | 2.18 - 2.23 |

Table 3: Known concentrations of common leafy vegetables, queried from Gemini 2025, normalized to mg/g of fresh weight.

### 5.5.1 Overall results

| Mean $\bar{x}$ | Standard Deviation $\text{std}(x)$ | Median $\tilde{x}$ | Range $\text{max}(x) - \text{min}(x)$ |
|----------------|------------------------------------|--------------------|---------------------------------------|
| 0.2761 mg/g    | 0.1978 mg/g                        | 0.1785 mg/g        | 0.6896 - 0.0646 mg/g                  |

Table 4: Descriptive statistics of the total chlorophyll concentration for all groups

Table 4 shows all descriptive statistics of the total chlorophyll concentration in fresh weight for all groups. We can use this information to put our group's results in context.

Based on the instructor's formula for the total chlorophyll concentration, our total chlorophyll concentration in fresh weight is 0.4838 mg/g. This belongs to the higher end of our value range, but is not an outlier. It indicates a result value fitting for our experiment. The high value makes sense, as Table 3 shows that brussels sprouts have a very high concentration of chlorophyll.

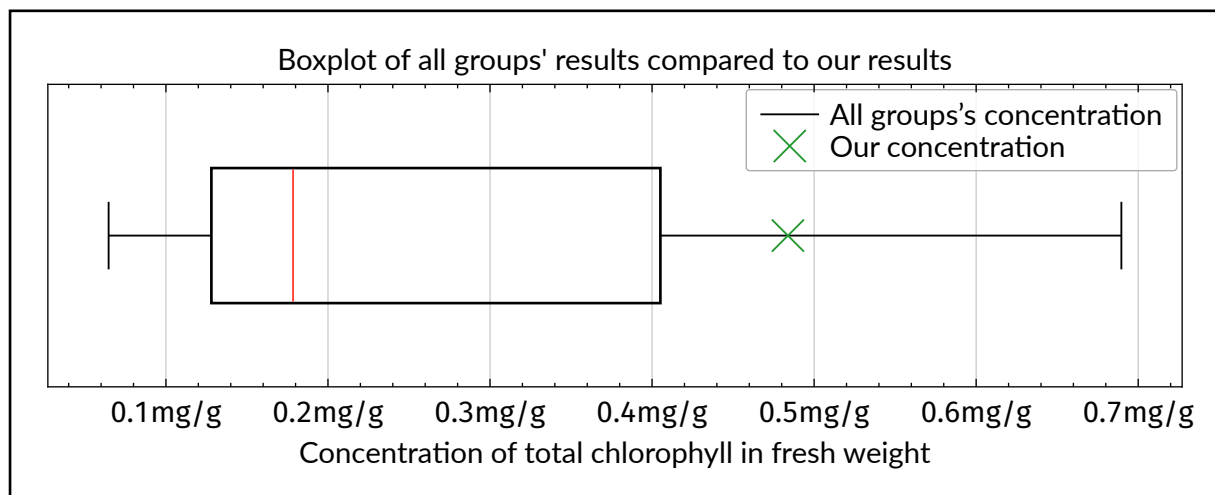


Figure 5: Boxplot of all groups' results compared to our results

Figure 5 shows a boxplot of all groups' results compared to our results. It demonstrated we lie between the third and fourth quartile of the data, and supports our claims above.

### 5.5.2 Results per source material

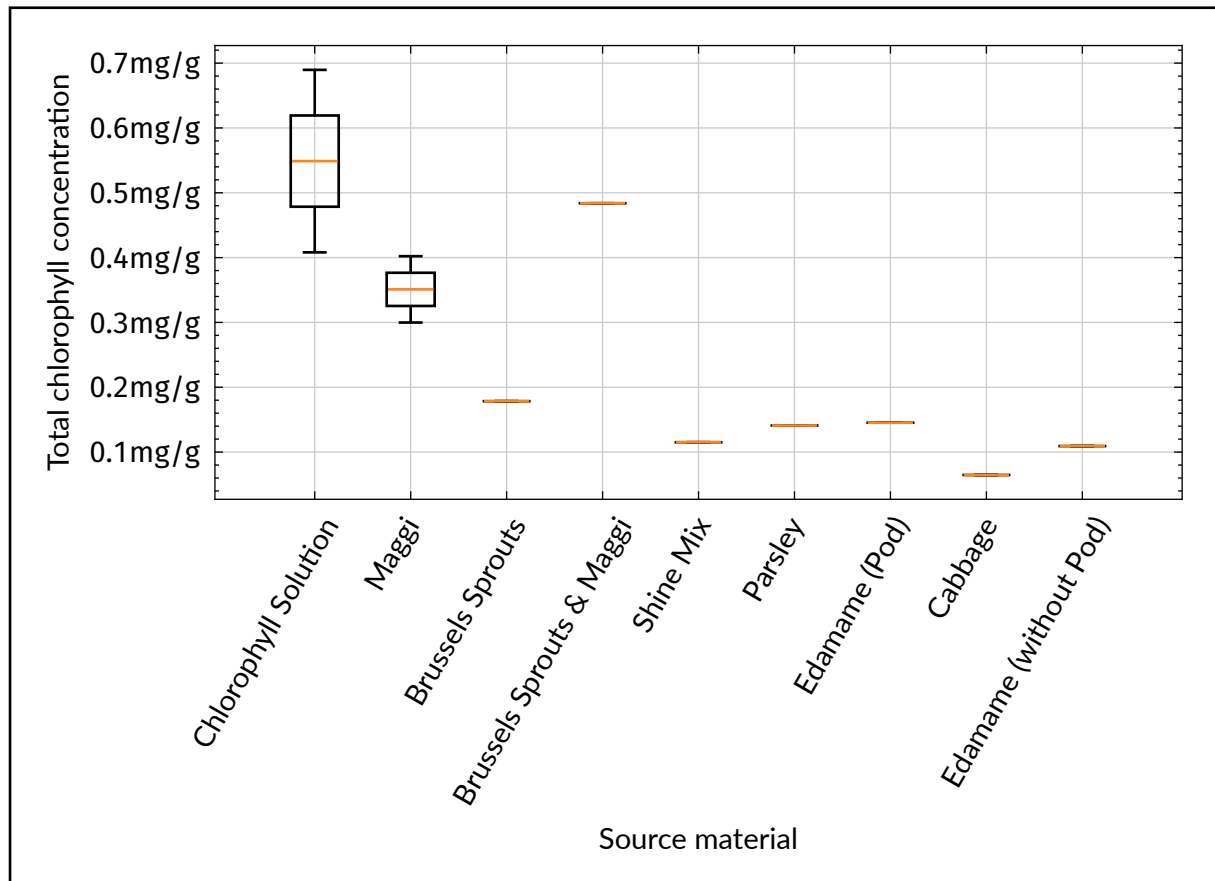


Figure 6: Boxplot of all groups' results per source material

Figure 6 shows interesting insights. Most of the source materials have only a singular entry and are therefore shown flat in the plot.

For the shine mix, parsley, edamame (with or without pods), and cabbage, the results are very similar. This is quite surprising, as Table 3 lists parsley with about  $15 \times$  higher concentration than we see in the boxplot.

The brussels sprouts and maggi mix has the highest concentration of the natural source materials (which excludes the chlorophyll solution). This should not be the case, as it should be somewhere in between the concentration for pure maggi herbs and brussels sprouts, yet it exceeds both.

One would expect the chlorophyll solution to have a very high concentration, as theoretically there should be a fresh weight concentration of 1mg/g (due to assuming 0.1% solution to be the fresh weight). We can see that trend reflected in the boxplot, even though the median is below the theoretical value.

This deviation is not unique to the chlorophyll solution, as virtually all natural source materials have a lower concentration than their expected value, based on Table 3.

#### 5.5.2.1 Explanation for the deviation

Several factors explain why our measured values may be lower than expected:

- **Incomplete extraction:** The acetone may not have extracted all chlorophyll from the plant cells. Different plants have different cell wall structures, making some harder to extract from than others.

- **Chlorophyll breakdown:** Chlorophyll is sensitive to light and heat. Some may have broken down during extraction and measurement, reducing the final concentration.
- **Measurement errors:** Small errors in weighing samples, measuring volumes, or reading the spectrophotometer can add up and affect the results.
- **Natural variation:** Real plants vary in chlorophyll content depending on age, growing conditions, and storage. The reference values are averages that may not match our specific samples.

We do not conclude the calculations or values to be catastrophically wrong, as even though there are wildly different raw data points, the overall end results of the calculation chain seem to settle in sensible values, as illustrated in Figure 7. Instead, we assume that error sources are small but consistent across all groups.

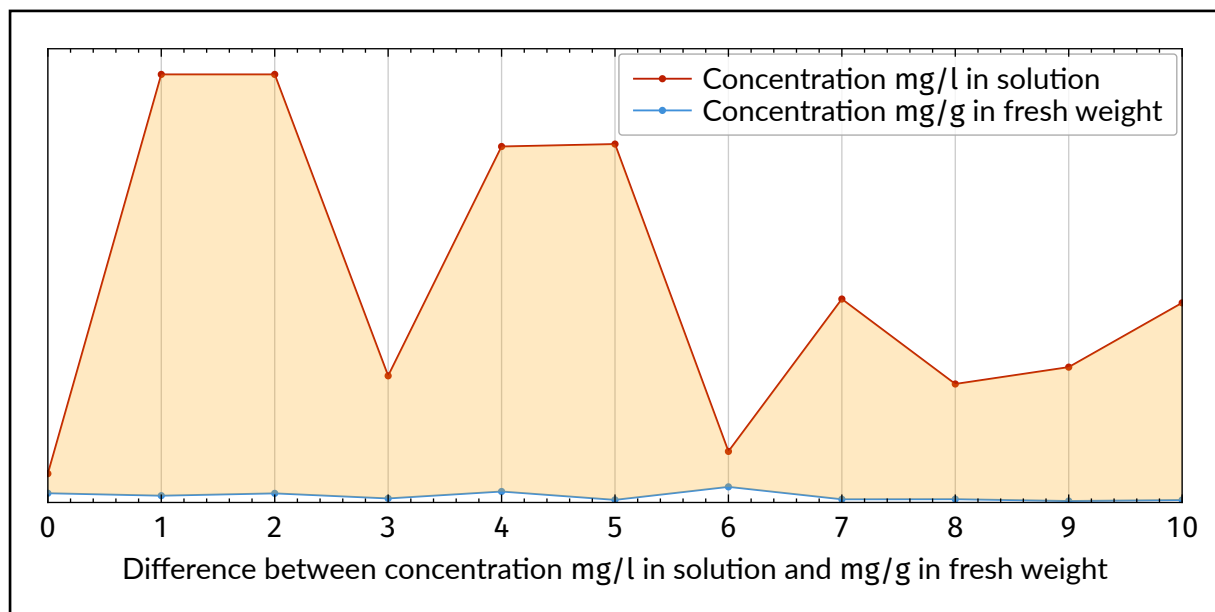


Figure 7: Difference between chlorophyll concentration in solution and in fresh weight for all groups, showing the diverge

| Data Points                        | Standard Deviation | Normalized Standard Deviation |
|------------------------------------|--------------------|-------------------------------|
| Concentration mg/l in solution     | $\pm 23.527$ mg/l  | $\pm 0.6645$                  |
| Concentration mg/g in fresh weight | $\pm 0.1978$ mg/g  | $\pm 0.7165$                  |

Table 5: Standard deviation of the data points for concentration in solution and in fresh weight

We can also compare the standard deviation of the soluted data points and the fresh weight data points. Table 5 shows that the compensation of the dilution factor works well, since we can effectively map all data points to a smaller more concrete range. This however is not perfect, as the *normalized*<sup>2</sup> standard deviation of the fresh weight data points is slightly higher than the standard deviation of the solution data points. It shows that when accounted for unit differences, the fresh weight data points are more scattered than the solution data points.

### 5.5.3 Comparison between fresh weight concentration sources

As mentioned in the introduction of Section 5.5, until now we have ignored the filled out concentrations on the results sheet, and instead recalculated using the instructor's formula based on the written down extinction coefficients. We will now compare these freshly recalculated

<sup>2</sup>normalized: data points divided by their common mean

concentrations with the concentrations filled out on the results sheets. For this purpose, we take the  $\text{Chl}_a$  and  $\text{Chl}_b$  concentrations filled out on the results sheet, add them together and compensate for the dilution factor to get the total chlorophyll concentration in fresh weight.

Afterwards, we compare the freshly recalculated total concentration with a total concentration calculated using the instructor's formula based on the 652nm extinction coefficient.

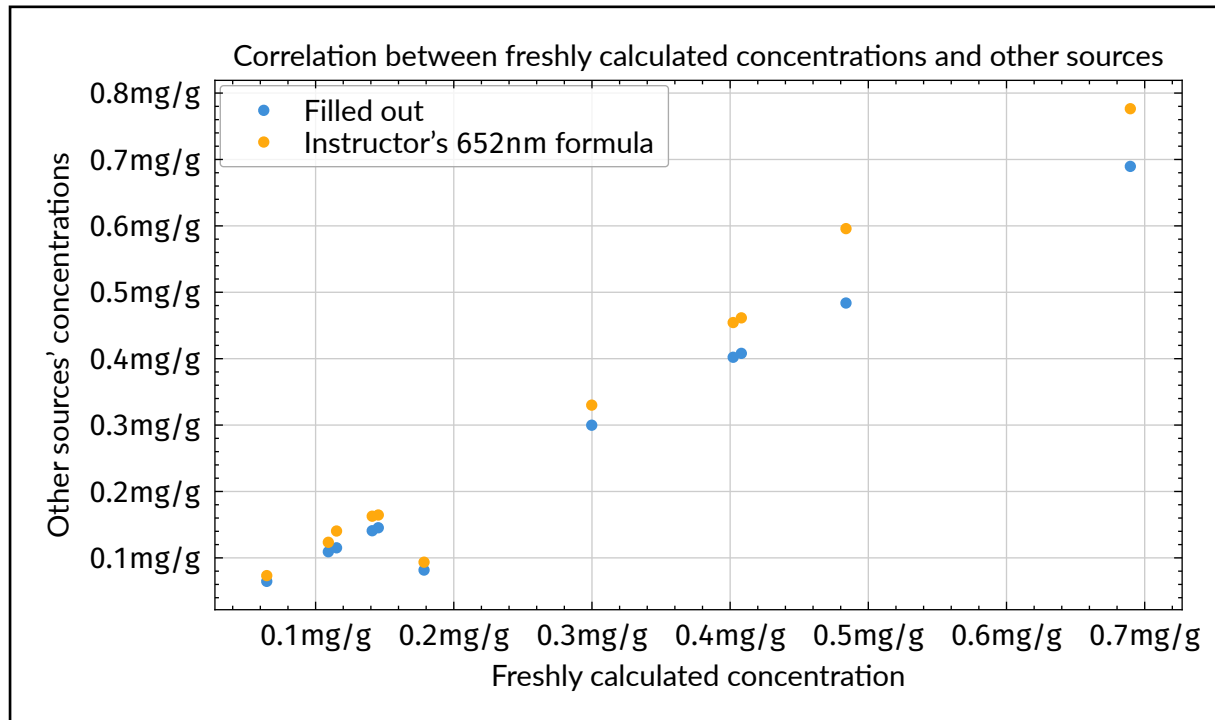


Figure 8: Comparison between freshly calculated concentrations and other sources using a scatter plot for correlation visualization, showing strong linear correlation between the sources

#### 5.5.3.1 Filled out & freshly calculated

There is almost no difference between the filled out and freshly calculated concentrations. Figure 9 shows that there is only one entry with a difference of  $\sim 1 \text{ mg/g}$  between the two values. This discrepancy is likely due to an error in writing down the concentration values on the results sheet, rather than a systematic calculation error, as the difference is isolated to a single data point and not significant enough to indicate a fundamental issue with the calculation methodology itself.

Furthermore, Figure 8 implies a strong linear correlation between the filled out and freshly calculated concentrations, with a correlation coefficient of 0.990048, which is very close to 1, supporting this observation. It shows, that there was little error in the group individual calculations, as the filled out concentrations are very close to the freshly calculated concentrations.

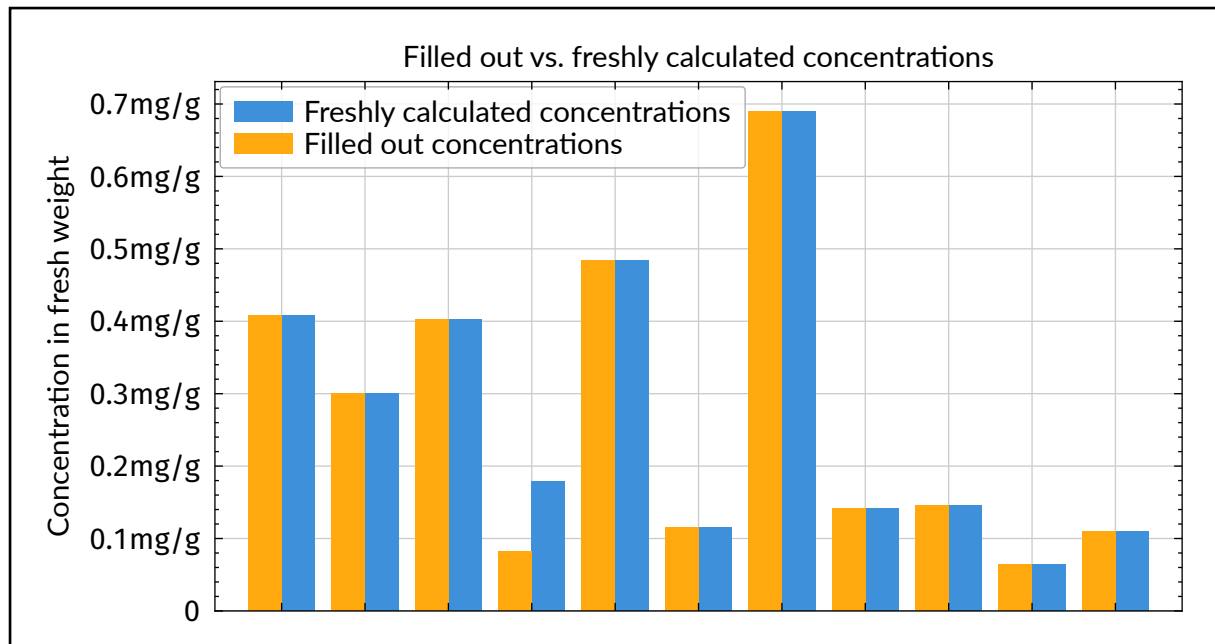


Figure 9: Comparison between filled out concentrations and freshly calculated concentrations

#### 5.5.3.2 Direct 652nm formula & freshly calculated

The instructor's formulas in Equation 6 for calculating the total chlorophyll concentration describe, how to use the 652nm extinction coefficient for that purpose. Figure 8 shows the correlation between the freshly calculated concentrations and the concentrations calculated using the 652nm instructor's formula. There we can also see a strong linear correlation between the two sources, with a correlation coefficient of 0.987914.

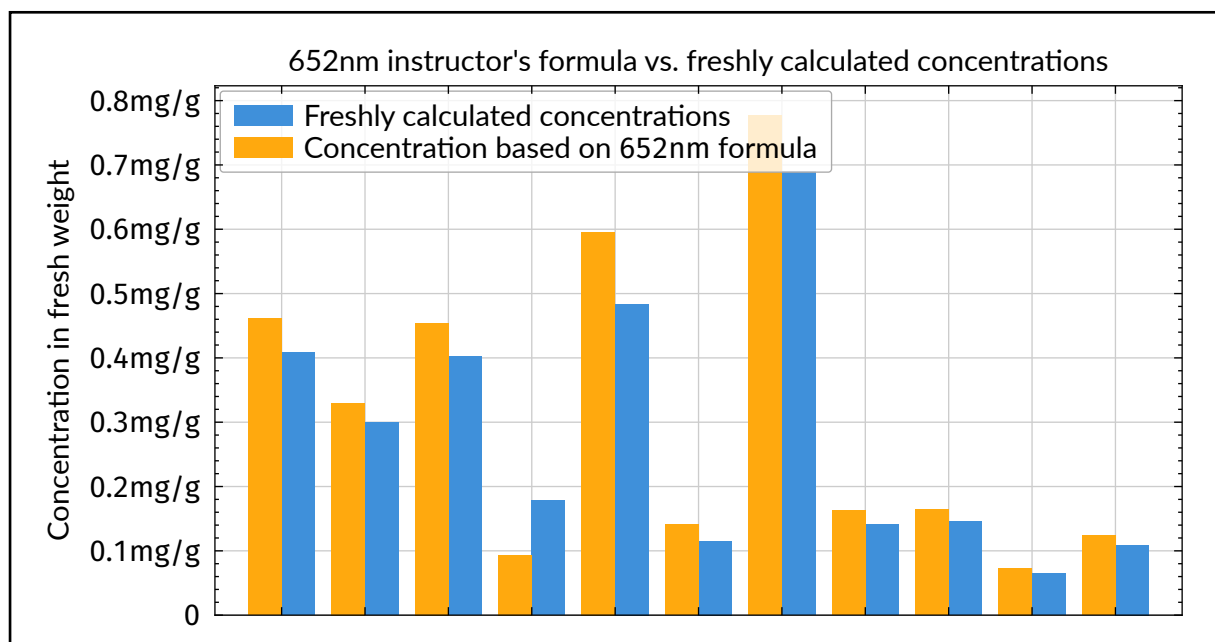


Figure 10: Comparison between 652nm instructor's formula and freshly calculated concentrations

Figure 10 shows that there is only a small difference between the two sources. Apart from the same outlier as in Section 5.5.3.1, there seems to be a positive offset in regards to the the 652nm formula. This offset seems to grow proportionally to the concentration and therefore classifies it as a fractional offset.

Interestingly enough, at closer observation of Figure 8, a strong correlation between the 652nm instructor's formula and the filled out concentrations is visible. Indeed, the correlation coefficient is 0.997967, which is practically equivalent to 1.

All of the above observations lead us to conclude, that the instructor's formula is very precise, but has a certain constant percentile accuracy offset. Lowering the factor of  $27.8 E_{652}$  by a nodge might result in a more accurate calculation (assuming the addition of  $\text{Chl}_a$  and  $\text{Chl}_b$  is the accurate baseline).

## Appendix

### Sources

- [1] J. S. Lindsey, M. Taniguchi, and H. Du, "PhotochemCAD | Chlorophyll a." Accessed: Dec. 01, 2025. [Online]. Available: <https://www.photochemcad.com/databases/common-compounds/chlorins-bacteriochlorins/chlorophyll-a>
- [2] J. S. Lindsey, M. Taniguchi, and H. Du, "PhotochemCAD | Chlorophyll b." Accessed: Dec. 01, 2025. [Online]. Available: <https://www.photochemcad.com/databases/common-compounds/chlorins-bacteriochlorins/chlorophyll-b>
- [3] H. K. Lichtenthaler, "Chlorophyll and Carotenoid Determination." Accessed: Dec. 05, 2025. [Online]. Available: [https://www.jkip.kit.edu/molbio/download/1A\\_Chlorophylls\\_Carotenoid\\_determin-HKLi\\_2010.pdf](https://www.jkip.kit.edu/molbio/download/1A_Chlorophylls_Carotenoid_determin-HKLi_2010.pdf)
- [4] Hendrik Poorter and Yvonne de Jong-Van Berkel, "Chlorophyll Extraction and Determination." Accessed: Dec. 09, 2025. [Online]. Available: <https://prometheusprotocols.net/function/tissue-chemistry/pigments/chlorophyll-extraction-and-determination/>
- [5] Australian Society of Plant Scientists, *Plants in Action*, 2nd ed. 2018. Accessed: Dec. 06, 2025. [Online]. Available: <https://rseco.org/content/122-chlorophyll-absorption-and-photosynthetic-action-spectra.html>
- [6] "Solvatochromism - Wikipedia." Accessed: Dec. 05, 2025. [Online]. Available: <https://en.wikipedia.org/wiki/Solvatochromism>
- [7] P. J. Pelletier and J. B. Caventou, "Recherches chimiques sur la matière verte des feuilles," *Annales de Chimie et de Physique*, vol. 9, 1818.
- [8] R. Willstätter and A. Stoll, *Untersuchungen über Chlorophyll*. Springer, 1913.
- [9] D. I. Arnon, "Copper enzymes in isolated chloroplasts. Polyphenoloxidase in *Beta vulgaris*," *Plant Physiology*, vol. 24, 1949, Accessed: Dec. 30, 2025. [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC437905/>
- [10] A. R. Wellburn, "The spectral determination of chlorophylls a and b, as well as total carotenoids, using various solvents with spectrophotometers of different resolution," *Journal of Plant Physiology*, vol. 144, 1994, doi: [https://doi.org/10.1016/S0176-1617\(11\)81192-2](https://doi.org/10.1016/S0176-1617(11)81192-2).
- [11] R. J. Porra, W. A. Thompson, and P. E. Kriedemann, "Determination of accurate extinction coefficients and simultaneous equations for assaying chlorophylls a and b extracted with four different solvents," *Biochimica et Biophysica Acta (BBA) - Bioenergetics*, vol. 975, 1989, doi: [https://doi.org/10.1016/S0005-2728\(89\)80347-0](https://doi.org/10.1016/S0005-2728(89)80347-0).
- [12] "Molar extinction coefficient - Wikipedia." Accessed: Dec. 29, 2025. [Online]. Available: [https://en.wikipedia.org/wiki/Molar\\_absorption\\_coefficient](https://en.wikipedia.org/wiki/Molar_absorption_coefficient)
- [13] "Quartz cuvette - Qvarz LLC." Accessed: Dec. 29, 2025. [Online]. Available: <https://qvarz.com/quartz-cuvette/>
- [14] M. Chazaux, C. Schiphorst, G. Lazzari, and S. Caffarri, "Precise estimation of chlorophyll a, b and carotenoid content by deconvolution of the absorption spectrum and new simulta-

neous equations for Chl determination," *The Plant Journal*, vol. 109, no. 6, pp. 1630–1648, 2022, doi: <https://doi.org/10.1111/tpj.15643>.

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